D 31198 **(Pages : 2)** **Name.....**

Reg. No.....

THIRD SEMESTER M.Sc. DEGREE (REGULAR/SUPPLEMENTARY) EXAMINATION, NOVEMBER 2022

(CBCSS)

Physics

PHY 3C 10—NUCLEAR AND PARTICLE PHYSICS

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

Section A

Answer **all** *questions.*
Each question carries weightage 1.

1. Sketch and explain the salient features of binding energy curve.
2. How the singlet and triplet potential existed in a nucleus ?
3. How did neutrino help in understanding of beta decay ?
4. Illustrate nuclear fusion process with example.
5. Explain the term quadrupole moment of a nucleus.
6. What is dead time in the GM counter ?
7. Classify particle based on interaction.
8. Briefly explain Grand Unified Theory.

(8 × 1 = 8 weightage)

Section B

Answer any **two** *questions.*
Each question carry weightage 5.

9. Outline the simple theory of deuteron structure using a square well potential of finite width and depth. Obtain the relation between the well parameters and binding energy. Show that the deuteron wave function is an admixture of S and D state.
10. Explain briefly the energetics of β decay reaction. Discuss the Fermi theory of β decay.
11. With a neat block diagram explain the working of a GM counter.
12. Discuss the fabrication and working of a controlled fusion reactor. State the application.

(2 × 5 = 10 weightage)

Turn over

Section C

Answer any **four** *questions.*
Each question carry weightage 3.

13. Determine the mass difference between two mirror nuclei which have N and Z differing by one unit and the same odd value of A.

14. Using semi empirical mass formula, find the most stable isobar for a nucleus having A = 43. Give coulomb energy coefficient a_c = 0.583 MeV and asymmetry energy coefficient a_{sym} = 19.3MeV.

15. Briefly compare the different type of gas detectors.

16. Explain Quark model for Spin Zero and Spin 3/2 hadrons.

17. Give an account of nuclear rotational energy levels.

18. Consider the following decay $^{17}\mathrm{F} \rightarrow {}^{17}\mathrm{O} + \beta^+ + \upsilon_e$ .Find the maximum kinetic energy of positron if the atomic masses of ^{17}F and ^{17}O are 17.0076 u and 17.0045 u respectively.

19. Draw a schematic diagram of nuclear fission reactor. Explain briefly a) Moderator b) Control rod.

(4 × 3 = 12 weightage)